

Threat Detection & SOC Dashboard Using Splunk Enterprise

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Tools: Splunk Enterprise 9.x | Oracle VirtualBox | Windows Server 2022
Framework: MITRE ATT&CK v14 | Techniques: T1110, T1110.001, T1110.003,
T1110.004

1. Project Overview

This project documents the design, build, and operation of a fully functional Security Operations Centre (SOC) home lab. The lab was built from scratch using Oracle VirtualBox, Windows Server 2022, and Splunk Enterprise to demonstrate real-world threat detection capabilities aligned to the MITRE ATT&CK framework.

1.1 Objectives

- Deploy and configure Splunk Universal Forwarder on Windows Server VMs
- Forward Windows Event Logs and Performance Monitor data to Splunk Enterprise
- Create MITRE ATT&CK-aligned detection alerts for brute force techniques (T1110)
- Simulate password spraying attacks and validate detection rules
- Build a live SOC dashboard with 8 detection and monitoring panels
- Document all findings in a structured SOC analyst report

1.2 Lab Environment

Component	Details
Hypervisor	Oracle VirtualBox
Host OS	Windows 11 (MOPELOLA-PC)
Guest OS	Windows Server 2022 (3 VMs)
SIEM	Splunk Enterprise 9.x
Log Forwarder	Splunk Universal Forwarder
Network Mode	Bridged Adapter (192.168.1.x)
Host IP	192.168.1.67
Receiving Port	9997
Framework	MITRE ATT&CK v14

2. Infrastructure Setup

2.1 Host Machine — ipconfig Output

The host machine's IPv4 address was identified as 192.168.1.67. This was used as the Splunk receiving server address in all Universal Forwarder configurations.

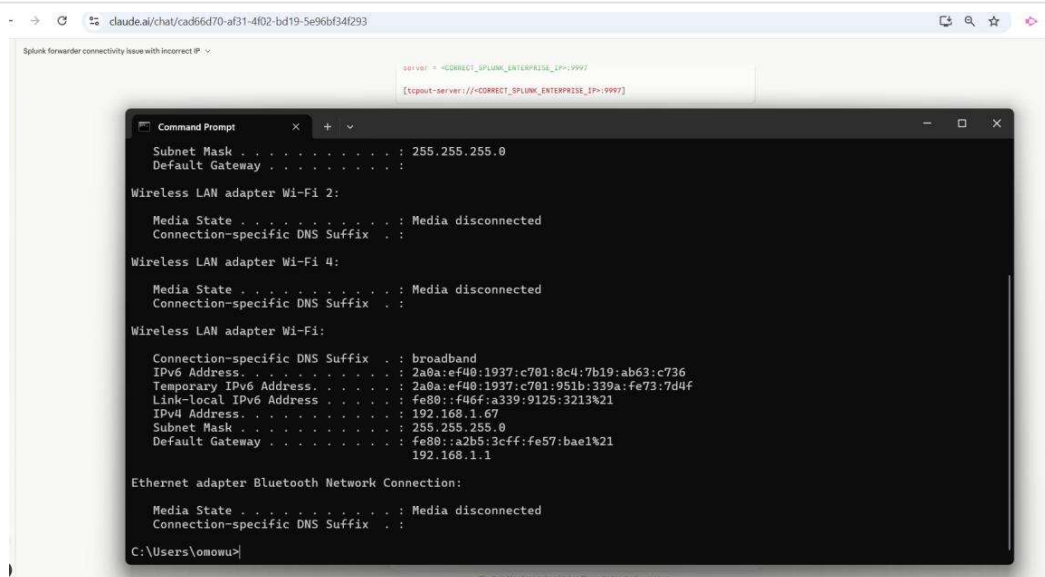


Figure 1: Host machine ipconfig — IPv4 address 192.168.1.67 on Wi-Fi adapter

2.2 VirtualBox Bridged Adapter Configuration

Initial connectivity failed because VMs were set to NAT mode. Changing to Bridged Adapter placed all VMs on the same 192.168.1.x subnet as the host, enabling direct communication.

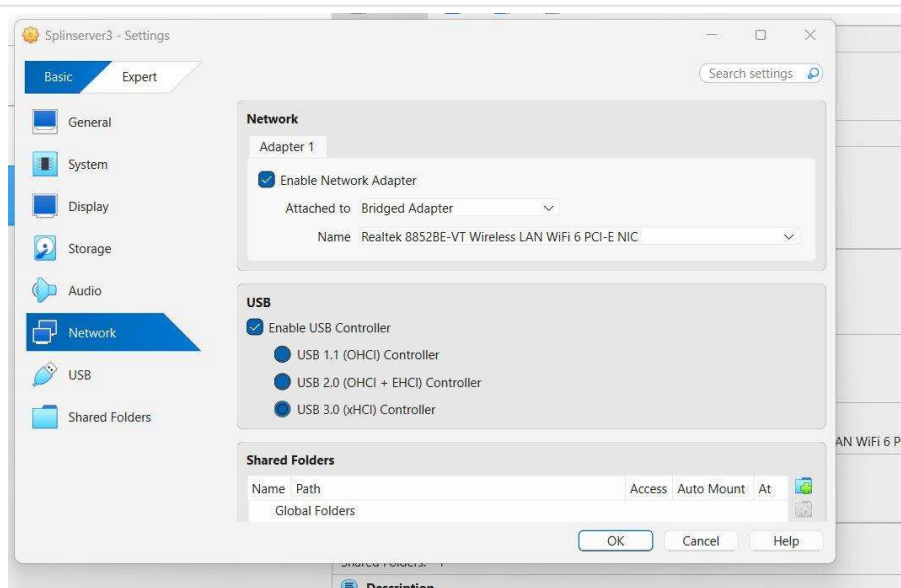


Figure 2: VirtualBox Bridged Adapter — Realtek WiFi 6 NIC selected

2.3 Connectivity Verification — Ping Tests

After fixing the VirtualBox network settings and adding the ICMPv4 firewall rule on the host, ping tests confirmed successful communication between all VMs and the host.

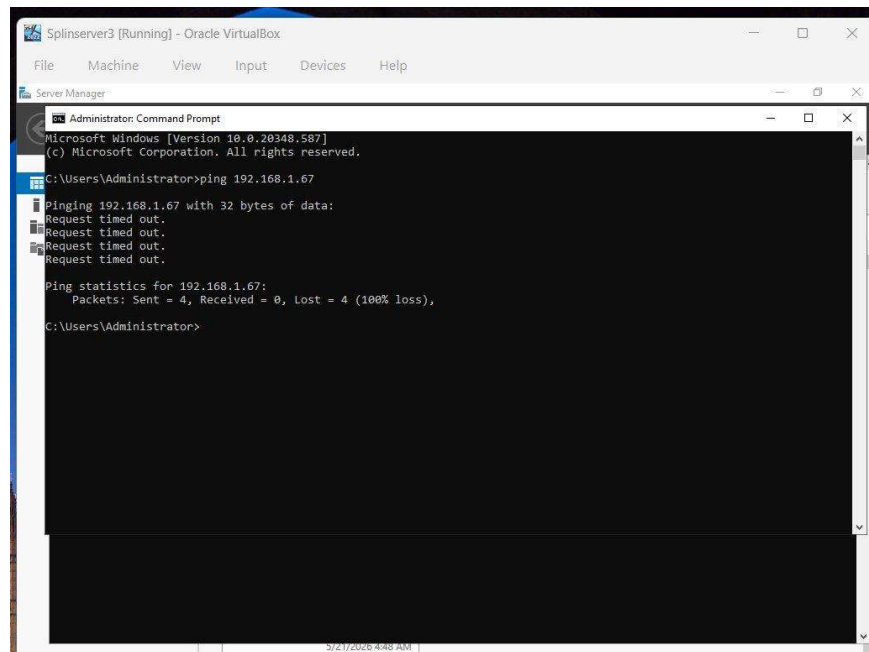
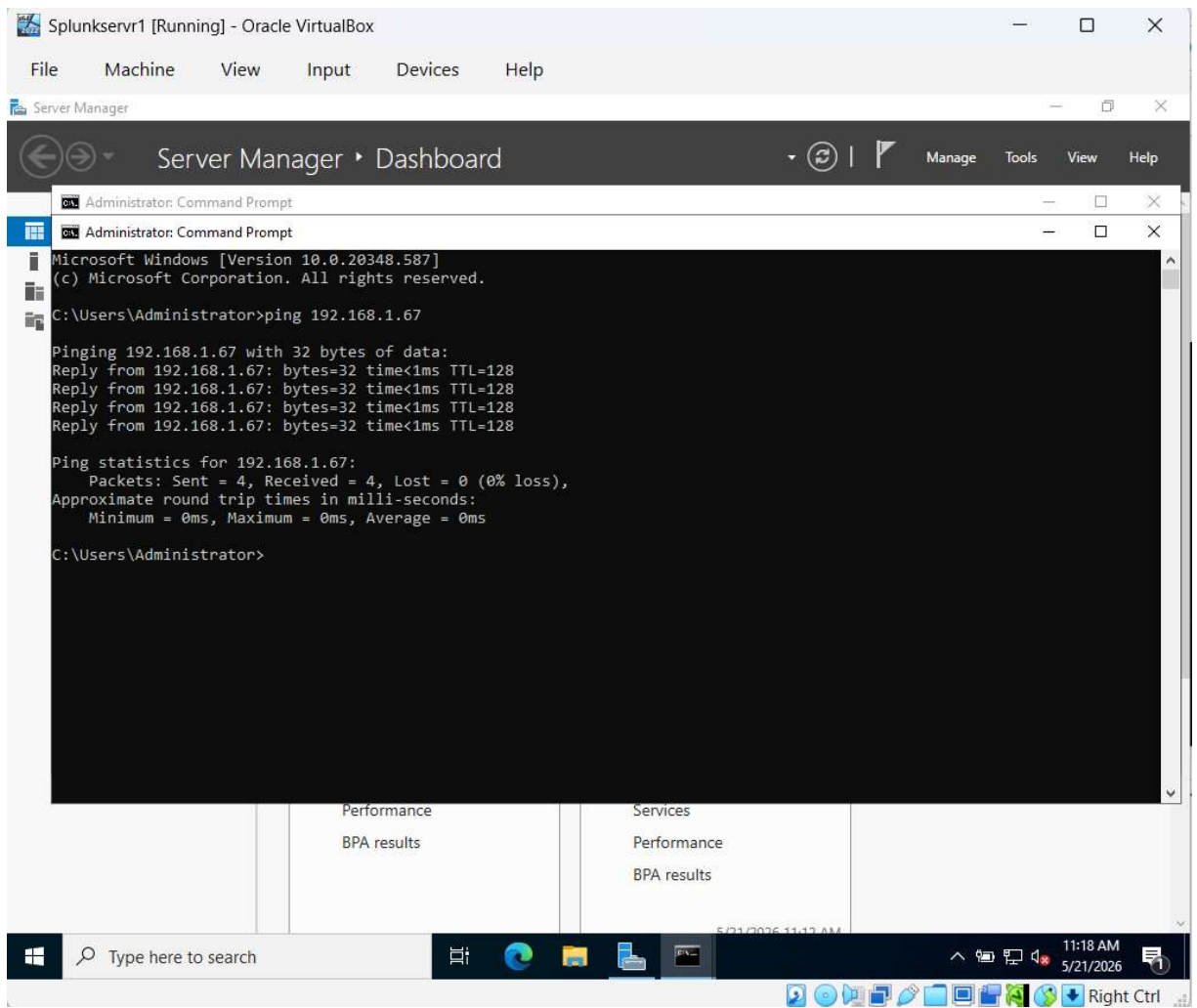


Figure 3: Ping failure — NAT mode before fix (Request timed out)



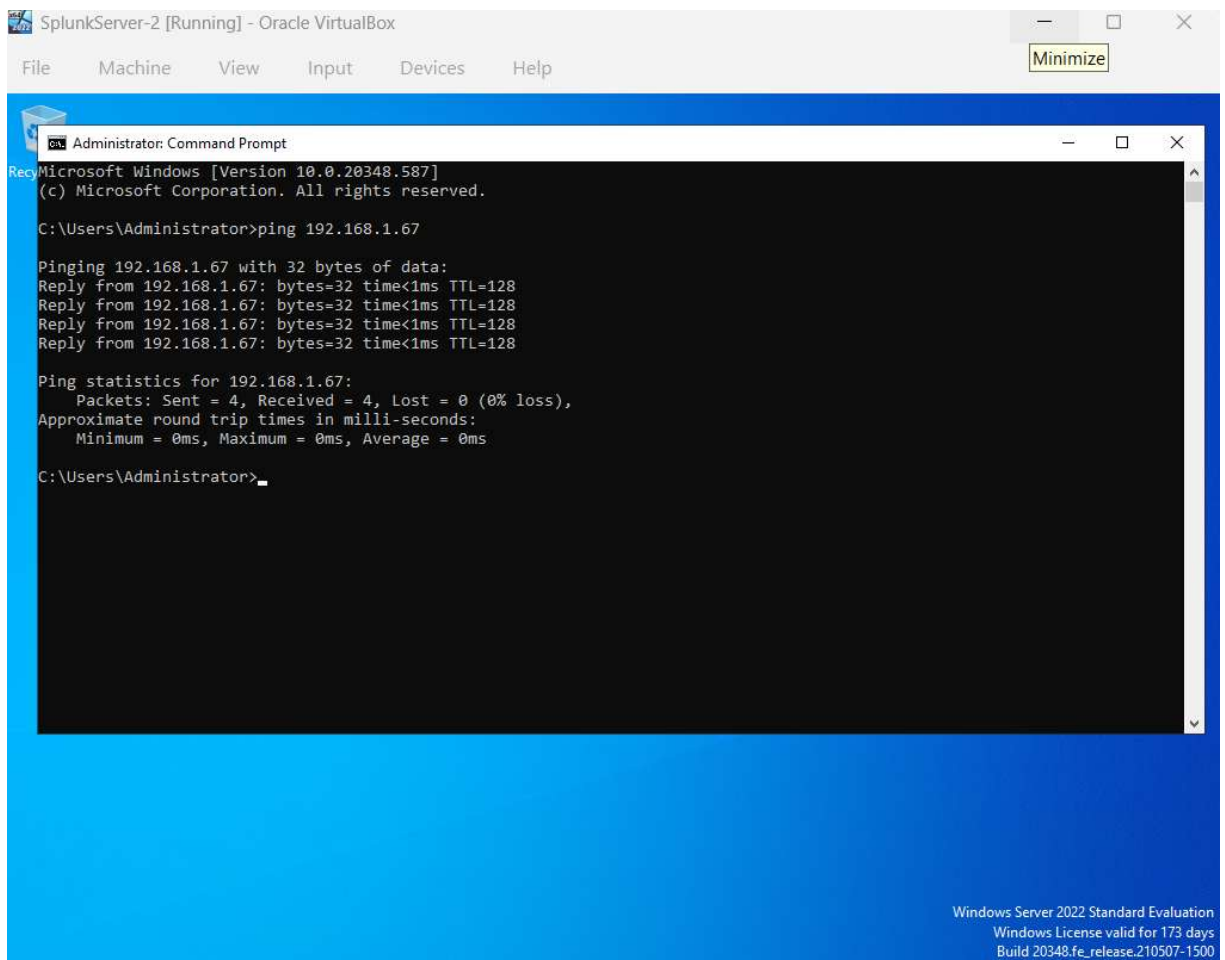


Figure 5: Ping success — after Bridged Adapter and firewall rule (0% packet loss)

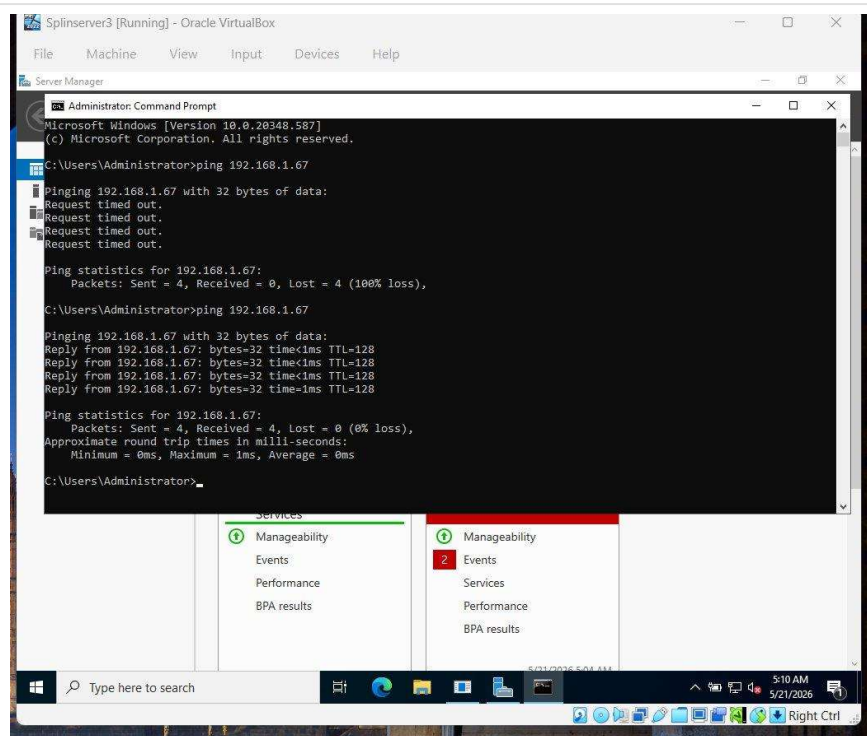


Figure 6: Ping success — after Bridged Adapter and firewall rule (0% packet loss)

2.4 Splunk Forwarder CLI Configuration

After navigating to the correct bin directory, the forward server was configured using the Splunk CLI on each VM:

```
cd "C:\Program Files\SplunkUniversalForwarder\bin"
splunk add forward-server 192.168.1.67:9997 -auth admin:yourpassword
splunk restart
```

Also with this method: [Completed Outbound Firewall Rule — Step by Step](#)

2.4.1 Step 1 — Open Windows Defender Firewall

Start → search "Windows Defender Firewall with Advanced Security"

→ Open as Administrator

Step 2 — Create Outbound Rule

Left panel → click "Outbound Rules"

→ Right panel → click "New Rule..."

Step 3 — Rule Type

Select: Program

Click: Next

Step 4 — Program Path

Select: "This program path"

Browse to:

C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe

Click: Next

Step 5 — Action

Select: Allow the connection

Click: Next

Step 6 — Profile

Check all three:

- ☒ Domain
- ☒ Private
- ☒ Public

Click: Next

Step 7 — Name the Rule

Name: Splunk Universal Forwarder – Outbound

Description: Allows splunkd.exe to forward logs to Splunk Enterprise on port 9997

Click: Finish

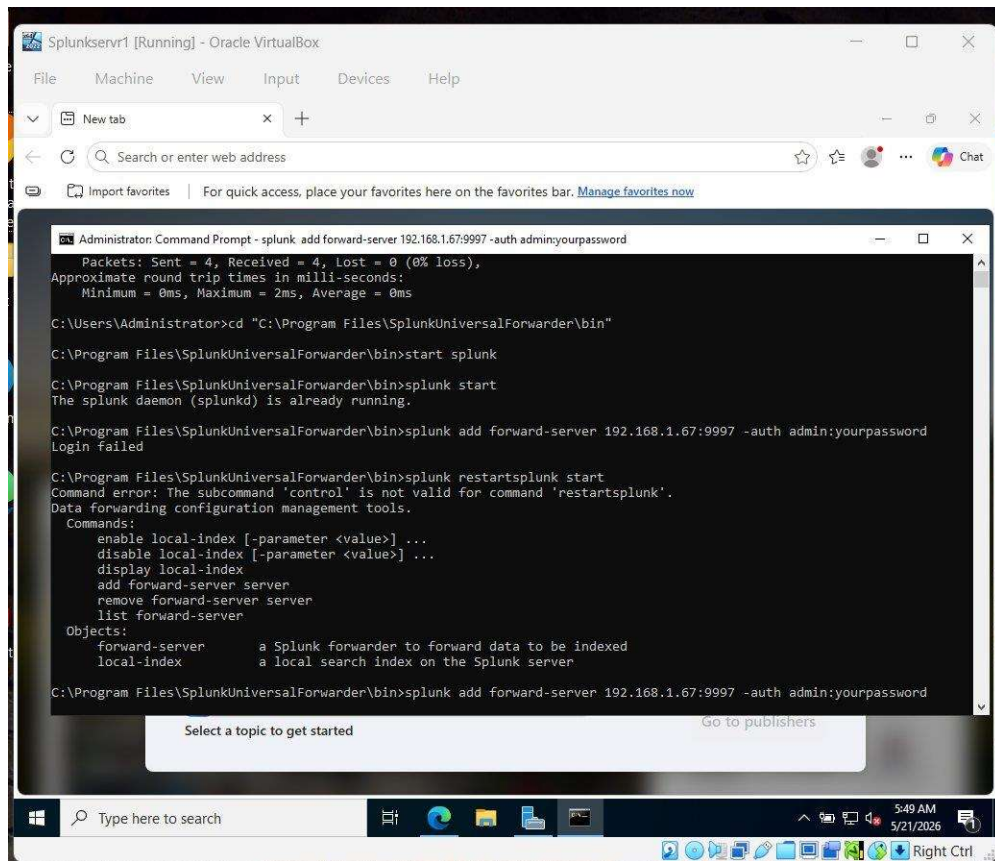


Figure 5: Splunk forwarder CLI — add forward-server command on VM

Full Firewall Rules carried out on both VM and Host

VMs

Rules	Direction	Program	Port	Action
Splunk UF Outbound	Out	Splunkd.exe	9997 TCP	Allow

Host

Rule	Direction	Port	Action
Splunk Receiver	In	9997 TCP	Allow
Allow ICMPv4	In	ICMP	Allow

3. Data Ingestion & Log Sources

3.1 All Hosts Connected — Data Summary

After all forwarders were correctly configured, Splunk Enterprise confirmed all 4 hosts actively forwarding data. Over 24,126 events were collected across all sourcetypes.

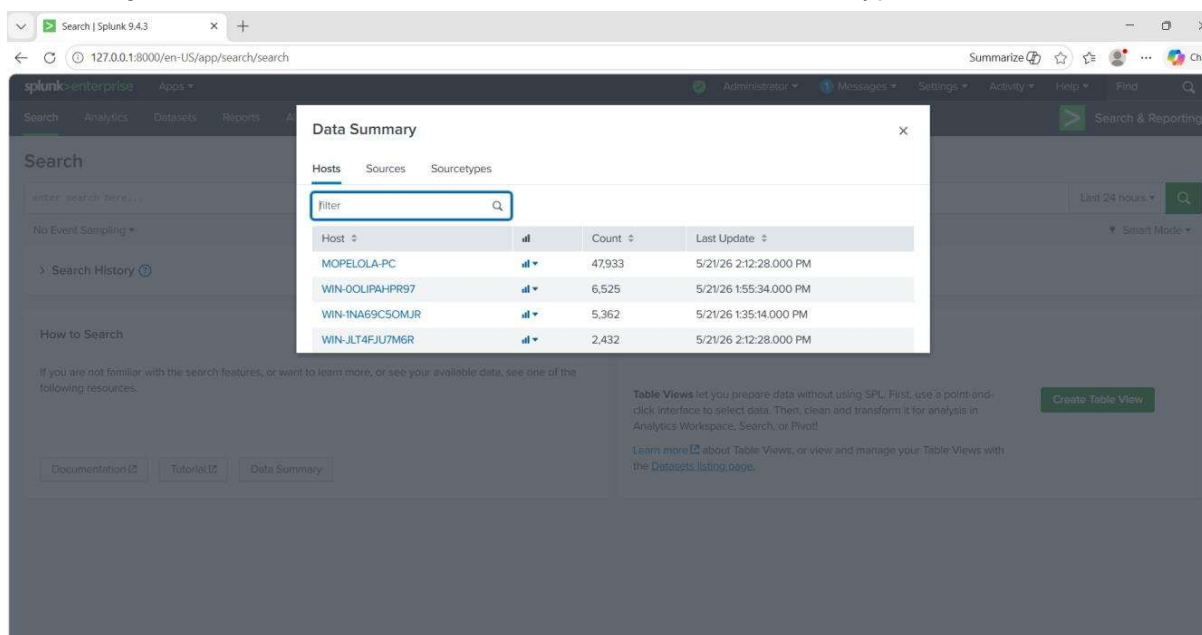


Figure 8: Splunk Data Summary — all 4 hosts connected and ingesting logs

3.2 Event Volume by Host and Sourcetype

Host	Sourcetypes	Events (24h)	Status
MOPELOLA-PC	WinEventLog: App, Security, Setup, System	3,954	Active
WIN-1NA69C5OMJR	WinEventLog + Perfmon: CPU, Memory, Network	7,337	Active
WIN-JLT4FJU7M6R	WinEventLog: Security, System + Perfmon	7,227	Active
WIN-00LIPAHPR97	WinEventLog + Perfmon: CPU, Memory, Network	5,608	Active

Figure 9: Event count by host and sourcetype — 24-hour window

3.3 Top Security Event Codes

EventCode	Description	Count	Risk Level
5379	Credential Manager read	25,943	Low — background
4624	Successful logon	2,988	Monitor
4672	Admin privilege assigned	2,817	Monitor
4625	Failed logon — T1110	190	HIGH
4907	Audit policy change	1,700	Medium
4648	Explicit credential use	17	Monitor — T1110.004

Events Patterns **Statistics (18)** Visualization

Show: 20 Per Page ☒ Format ☒ Preview: On

host	Account_Name	count
MOPELOLA-PC	MOPELOLA-PCS	185
MOPELOLA-PC	SYSTEM	183
WIN-BOLIPAHPR97	WIN-BOLIPAHPR97\$	83
WIN-BOLIPAHPR97	SYSTEM	81
WIN-1NA69CSOMJR	WIN-1NA69CSOMJR\$	74
WIN-1NA69CSOMJR	SYSTEM	72
WIN-JLT4FJUTM6R	WIN-JLT4FJUTM6R\$	62
WIN-JLT4FJUTM6R	SYSTEM	56
MOPELOLA-PC	Mopelola	2
WIN-BOLIPAHPR97	Administrator	2
WIN-1NA69CSOMJR	Administrator	2
WIN-JLT4FJUTM6R	DMN-1	2
WIN-JLT4FJUTM6R	-	1
WIN-JLT4FJUTM6R	Administrator	1
WIN-JLT4FJUTM6R	LOCAL SERVICE	1
WIN-JLT4FJUTM6R	NETWORK SERVICE	1
WIN-JLT4FJUTM6R	UNFD-0	1
WIN-JLT4FJUTM6R	UNFD-1	1

Figure 10: Top 10 security event codes — EventCode 5379 dominant at 62.5%

4. MITRE ATT&CK Detection Rules

4.1 Framework Coverage

MITRE ID	Technique	Detection Method	Alert Status
T1110	Brute Force	EventCode 4625 threshold >= 5	Enabled
T1110.001	Password Guessing	4625 per account < 3 unique accounts	Enabled
T1110.003	Password Spraying	dc(Account_Name) >= 3 in 5-min window	Enabled
T1110.004	Credential Stuffing	EventCode 4648 count >= 5	Enabled

4.2 Brute Force Alert — T1110

```
index=* sourcetype=WinEventLog:Security EventCode=4625
| stats count by host, Account_Name
| where count >= 5 | sort -count
```

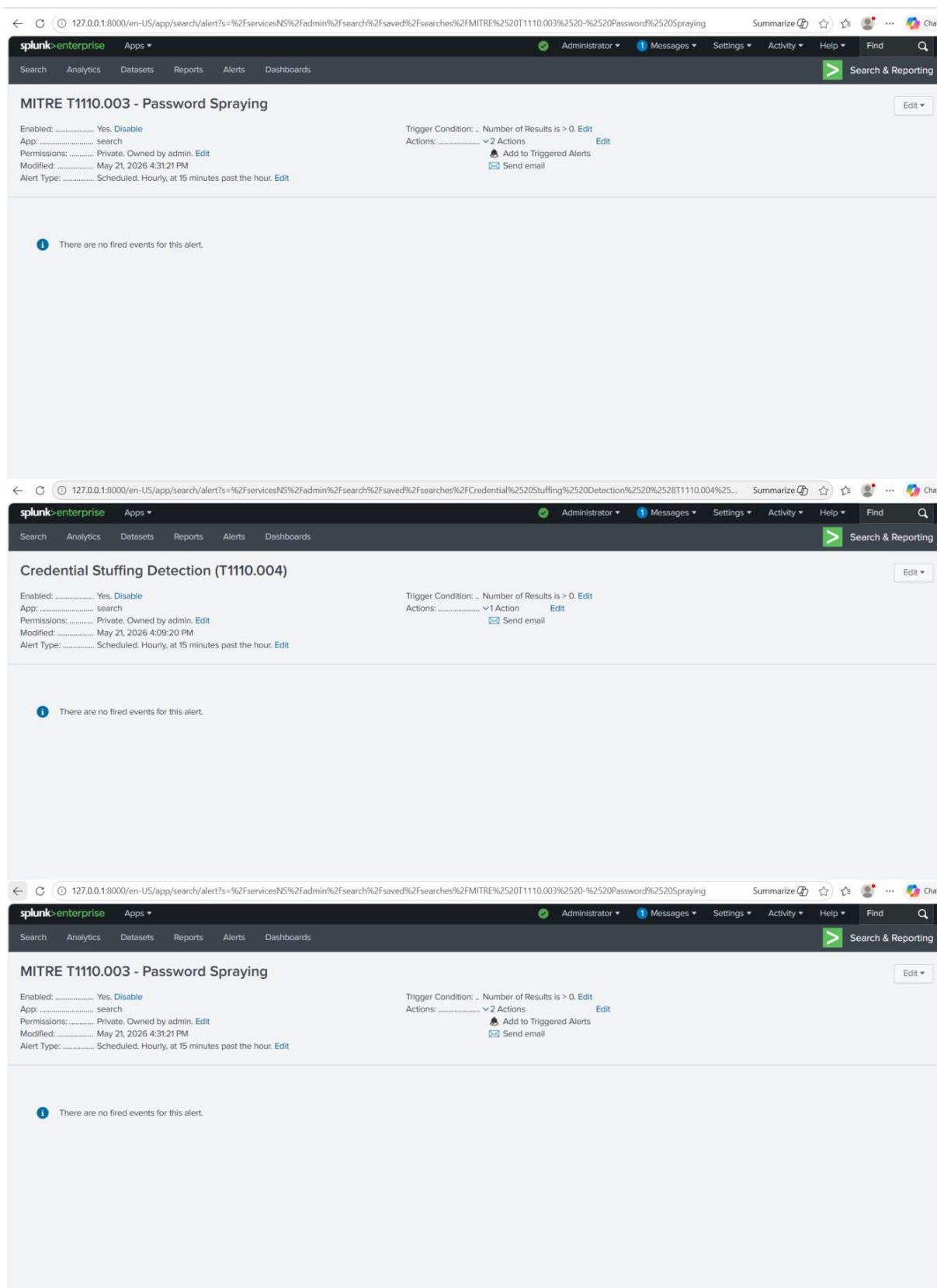


Figure 10: Save As Alert dialog — T1110 Brute Force Detection configuration

4.3 Password Spraying Detection — T1110.003

```
index=* sourcetype=WinEventLog:Security EventCode=4625
| bucket _time span=5m
| stats dc(Account_Name) as unique_accounts, count by _time, host
| where unique_accounts >= 3 AND count >= 10
| eval
Severity=case(count>=20,"CRITICAL",count>=15,"HIGH",count>=10,"MEDIUM",true(),"LOW")
| eval MITRE="T1110.003 - Password Spraying"
```

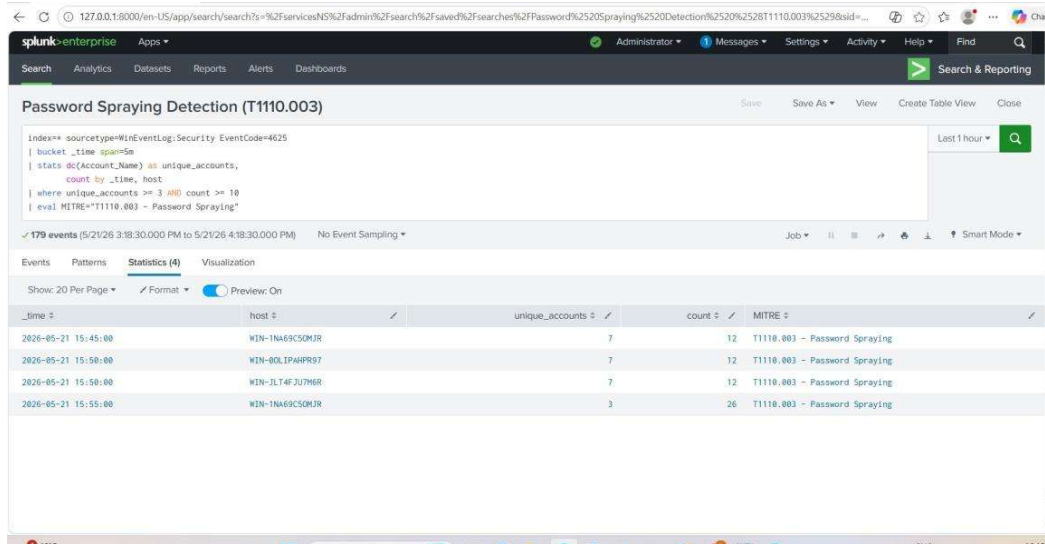


Figure 10: T1110.003 detection — CRITICAL and MEDIUM results across 3 VMs

4.4 All Alerts Configured and Enabled

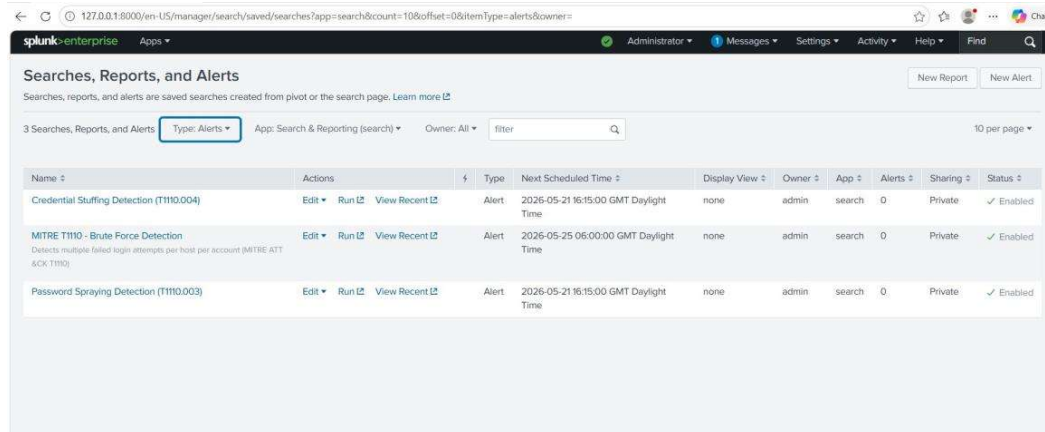
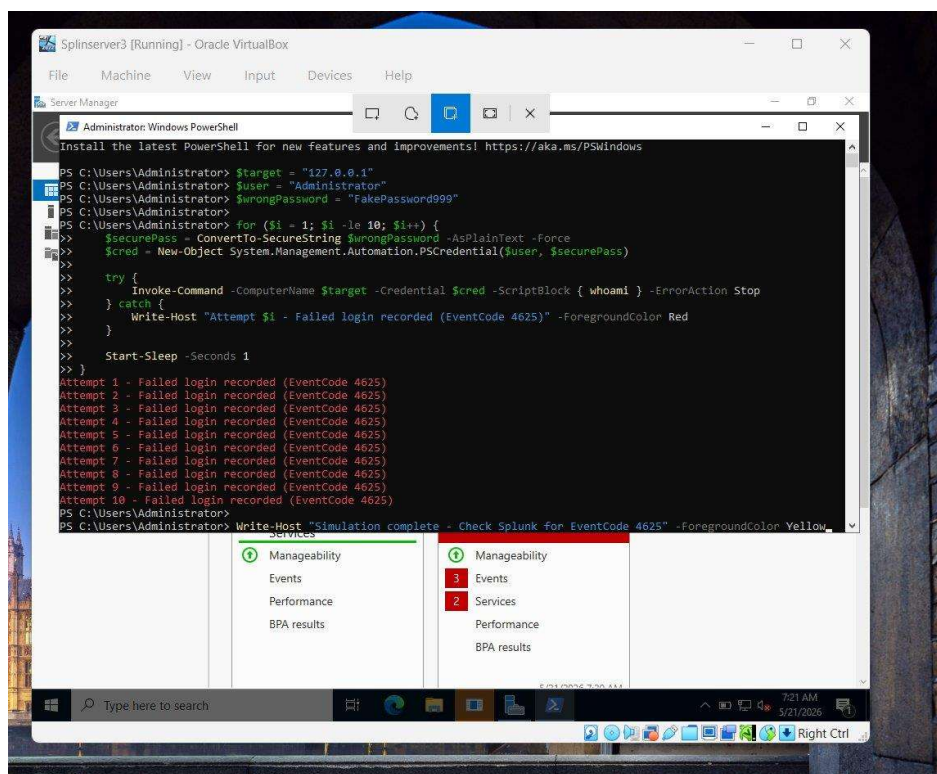


Figure 11: Settings — 3 MITRE alerts enabled with email action

5. Attack Simulation & Results

5.1 Brute Force Simulation — PowerShell Script

A PowerShell loop was executed on the VMs to generate 10 consecutive failed login attempts (EventCode 4625). This simulated a real brute force attack against the Administrator account.



```
PS C:\Users\Administrator> $target = "127.0.0.1"
PS C:\Users\Administrator> $user = "Administrator"
PS C:\Users\Administrator> $wrongPassword = "TakePassword999"
PS C:\Users\Administrator>
PS C:\Users\Administrator> for ($i = 1; $i -le 10; $i++) {
>> $securePass = ConvertTo-SecureString $wrongPassword -AsPlainText -Force
>> $cred = New-Object System.Management.Automation.PSCredential($user, $securePass)
>> try {
>> Invoke-Command -ComputerName $target -Credential $cred -ScriptBlock { whoami } -ErrorAction Stop
>> } catch {
>> Write-Host "Attempt $i - Failed login recorded (EventCode 4625)" -ForegroundColor Red
>> }
>> Start-Sleep -Seconds 1
>> }
Attempt 1 - Failed login recorded (EventCode 4625)
Attempt 2 - Failed login recorded (EventCode 4625)
Attempt 3 - Failed login recorded (EventCode 4625)
Attempt 4 - Failed login recorded (EventCode 4625)
Attempt 5 - Failed login recorded (EventCode 4625)
Attempt 6 - Failed login recorded (EventCode 4625)
Attempt 7 - Failed login recorded (EventCode 4625)
Attempt 8 - Failed login recorded (EventCode 4625)
Attempt 9 - Failed login recorded (EventCode 4625)
Attempt 10 - Failed login recorded (EventCode 4625)
PS C:\Users\Administrator> Write-Host "Simulation complete - Check Splunk for EventCode 4625" -ForegroundColor Yellow
```

Figure 12: PowerShell simulation — 10 failed logins generated on Splinserver3

5.2 Password Spraying Results — T1110.003 Confirmed

The T1110.003 detection rule successfully identified simultaneous password spraying across 3 VMs. The most severe detection showed 26 attempts across 3 unique accounts in a single 5-minute window — classified as CRITICAL.

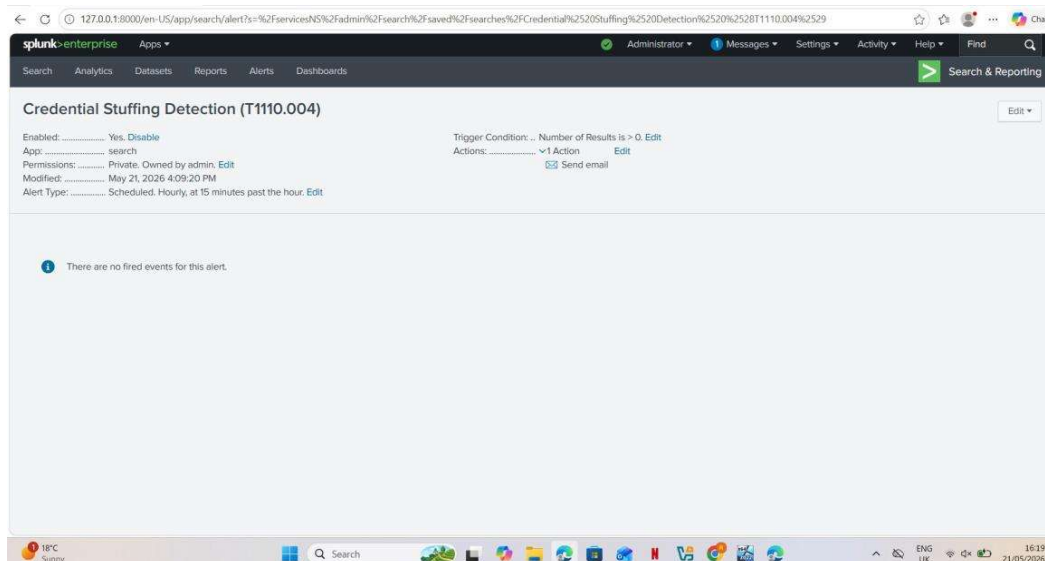
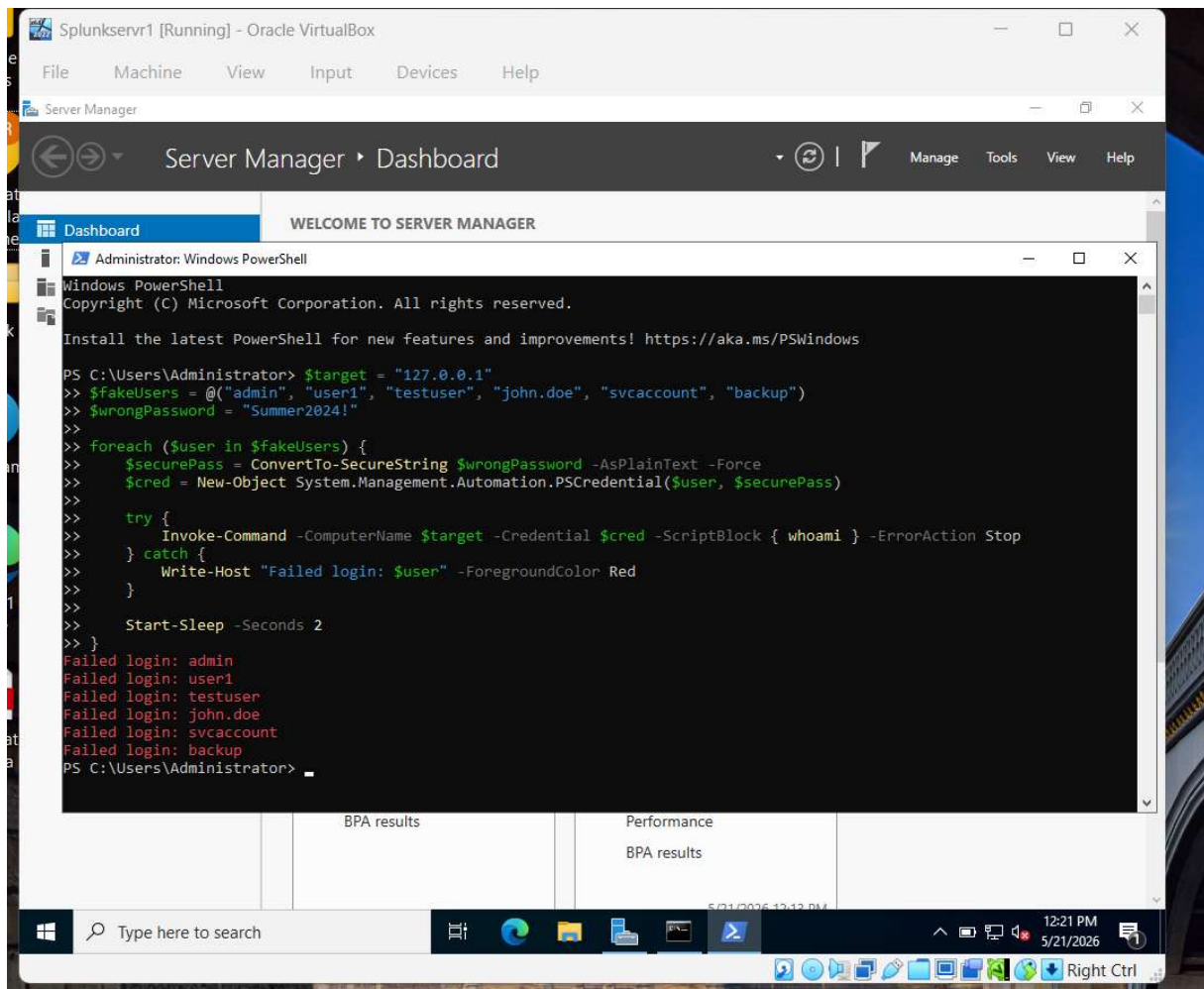


Figure 14: CRITICAL — WIN-1NA69C5OMJR 26 attempts; 3 VMs detected simultaneously

5.3 Failed Login Analysis — Targeted Accounts

Account	Failed Attempts	Risk Assessment
"-" (anonymous/null)	167	High — network-based attack pattern
Administrator	150	High — privileged account targeted
user1 / testuser / svcaccount	6 each	Medium — spray simulation accounts
john.doe / backup / admin	6 each	Medium — spray simulation accounts
Mopelola / MOPELOLA-PC\$	4 each	Low — host accounts

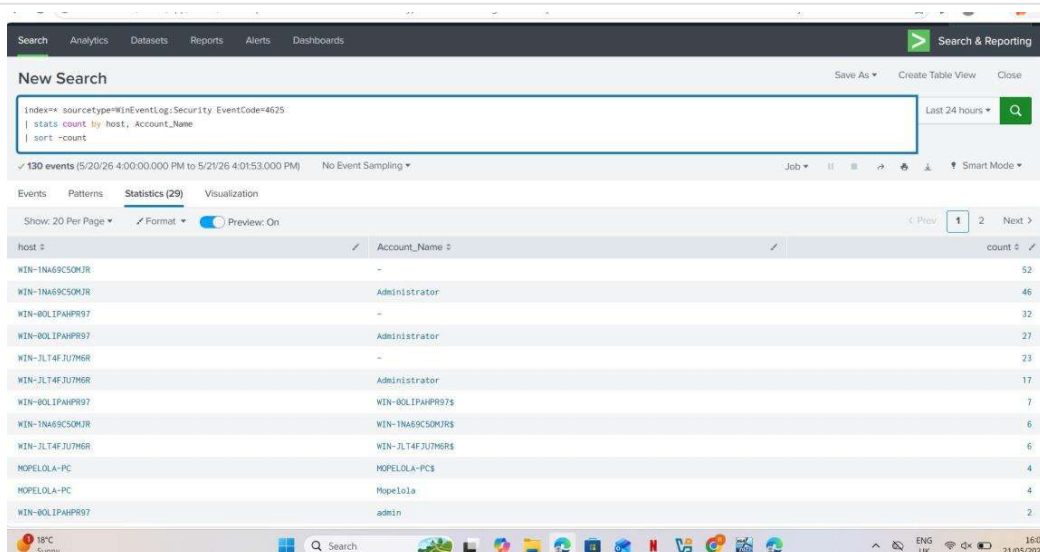


Figure 15: Failed logins by host and account — Administrator most targeted

5.4 Post-Attack Verification — No Compromise

A post-attack check confirmed no successful logins (EventCode 4624) occurred for targeted accounts immediately after the simulation. The attack did not result in credential compromise.

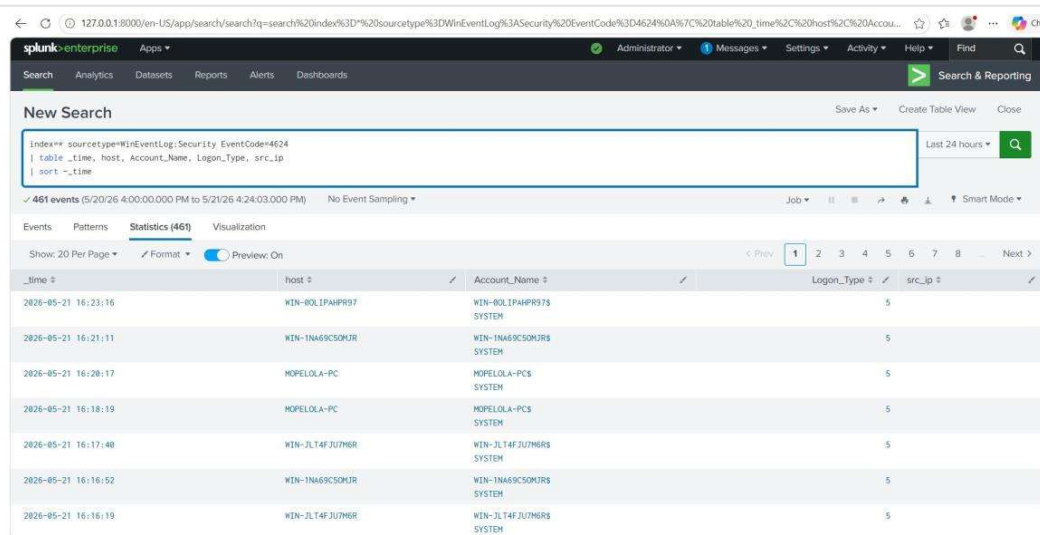
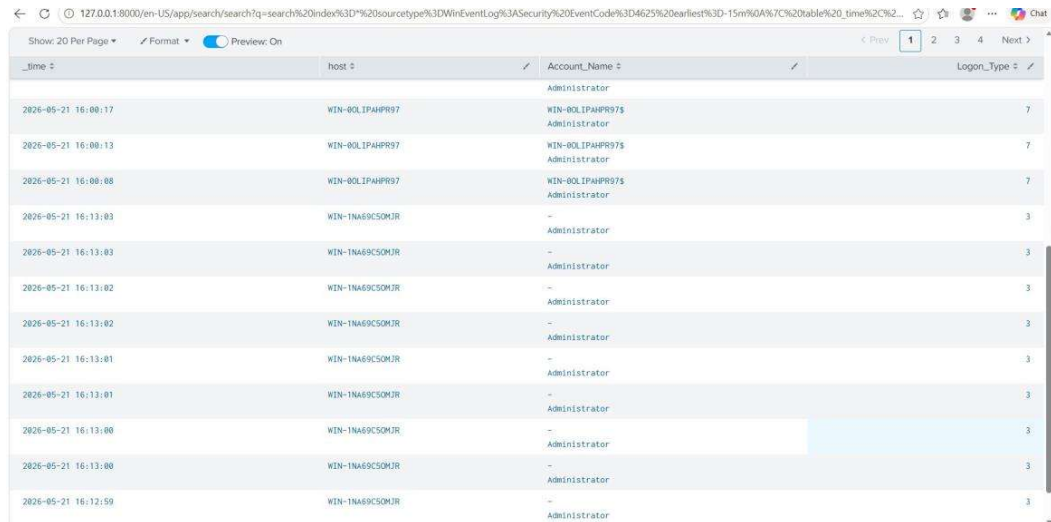


Figure 16: Post-attack check — only SYSTEM (Logon Type 5) logins present

6. SOC Analyst Investigation

6.1 Live Failed Login Feed — 15-Minute Window

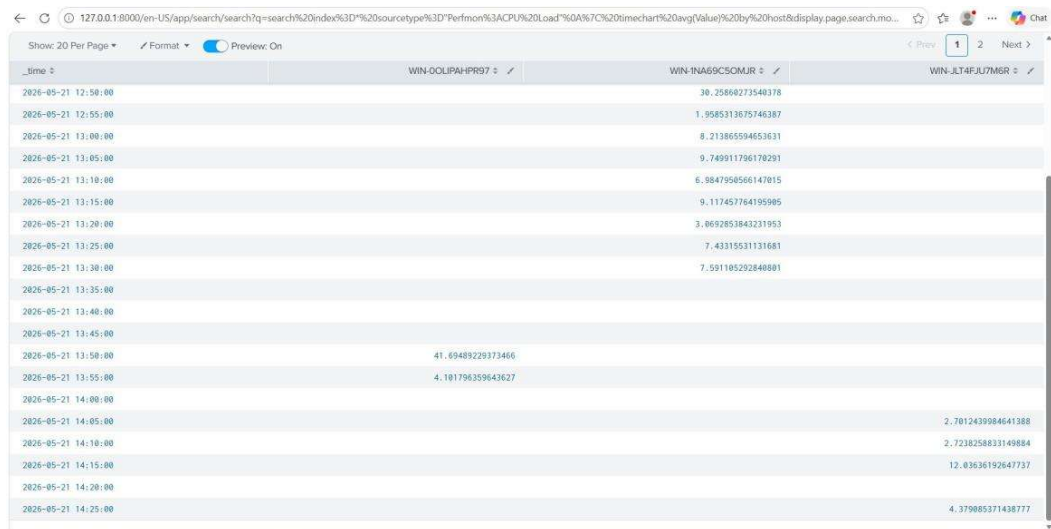


_time	host	Account_Name	Logon_Type
2026-05-21 16:00:17	WIN-00LIPAHPR97	Administrator	7
2026-05-21 16:00:13	WIN-00LIPAHPR97	Administrator	7
2026-05-21 16:00:08	WIN-00LIPAHPR97	Administrator	7
2026-05-21 16:13:03	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:03	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:02	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:02	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:01	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:01	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:00	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:13:00	WIN-1NA69C5OMJR	Administrator	3
2026-05-21 16:12:59	WIN-1NA69C5OMJR	Administrator	3

Figure 17: Real-time failed login feed — WIN-1NA69C5OMJR and WIN-00LIPAHPR97 targeted

6.2 Performance Monitoring — CPU Load

Perfmon data from VMs showed average CPU load of 2–12% during the simulation period — confirming the attack script ran without overwhelming system resources.



_time	WIN-00LIPAHPR97	WIN-1NA69C5OMJR	WIN-JLT4FJU7M8R
2026-05-21 12:58:00		30.25868273540378	
2026-05-21 12:55:00		1.9585313675746387	
2026-05-21 13:00:00		8.213865594653631	
2026-05-21 13:05:00		9.749911796178291	
2026-05-21 13:10:00		6.9847958566147015	
2026-05-21 13:15:00		9.117457764195985	
2026-05-21 13:20:00		3.8692853843231953	
2026-05-21 13:25:00		7.43315531131681	
2026-05-21 13:30:00		7.591185292848801	
2026-05-21 13:35:00			
2026-05-21 13:40:00			
2026-05-21 13:45:00			
2026-05-21 13:50:00	41.89489229373466		
2026-05-21 13:55:00	4.181796359643627		
2026-05-21 14:00:00			
2026-05-21 14:05:00			2.7612439984641388
2026-05-21 14:10:00			2.7238258833149884
2026-05-21 14:15:00			12.83636192647737
2026-05-21 14:20:00			
2026-05-21 14:25:00			4.379865371438777

Figure 18: CPU Load timechart — average usage across 3 VMs during simulation

6.3 Available Memory

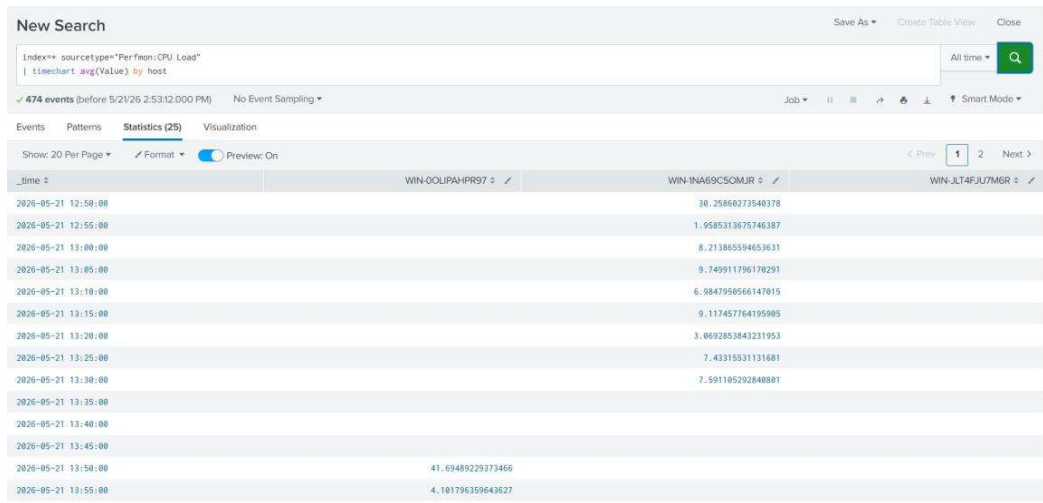


Figure 19: Available Memory — approximately 6.7GB on monitored VMs

7. SOC Dashboard

7.1 Dashboard Overview

A live MITRE ATT&CK SOC Dashboard was built in Splunk Enterprise with 8 panels covering event metrics, MITRE detections, bar charts, and a timeline. The dashboard auto-refreshes every 5 minutes and sends email alerts on threshold breach.

Panel	Visualization	Purpose
Total Events (24h)	Single Value	Confirms all forwarders active
Failed Logins — T1110	Single Value	Primary brute force indicator
Total Failed Logins	Single Value	Cumulative attack volume
Failed Logins by Host	Bar Chart	Most targeted host identification
Event Volume by Host	Bar Chart	Baseline activity per host
MITRE T1110.003 Detections	Statistics Table	Password spraying detections
Failed Logins 10-min Windows	Statistics Table	Burst detection with severity
Top Security Event Codes	Pie Chart	Event composition overview
Event Timeline by Host	Line Chart	Attack timeline visualisation

7.2 Dashboard — Page 1 (Metrics & Bar Charts)

Total events 24,126 — Failed logins T1110: 190 — Event volume and failed logins by host bar charts:

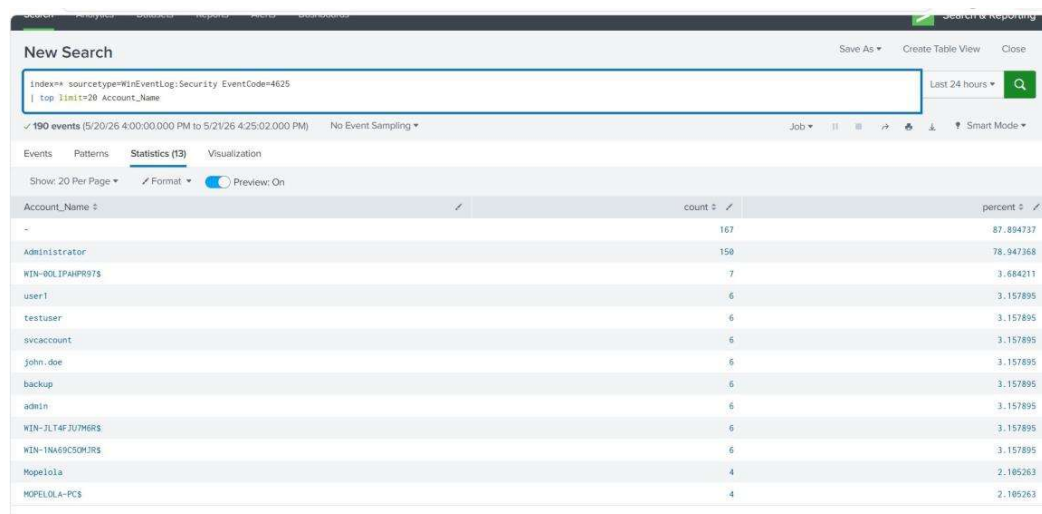


Figure 20: Dashboard Page 1 — metrics, event volume and failed logins bar charts

7.3 Dashboard — MITRE T1110.003 Detection Table

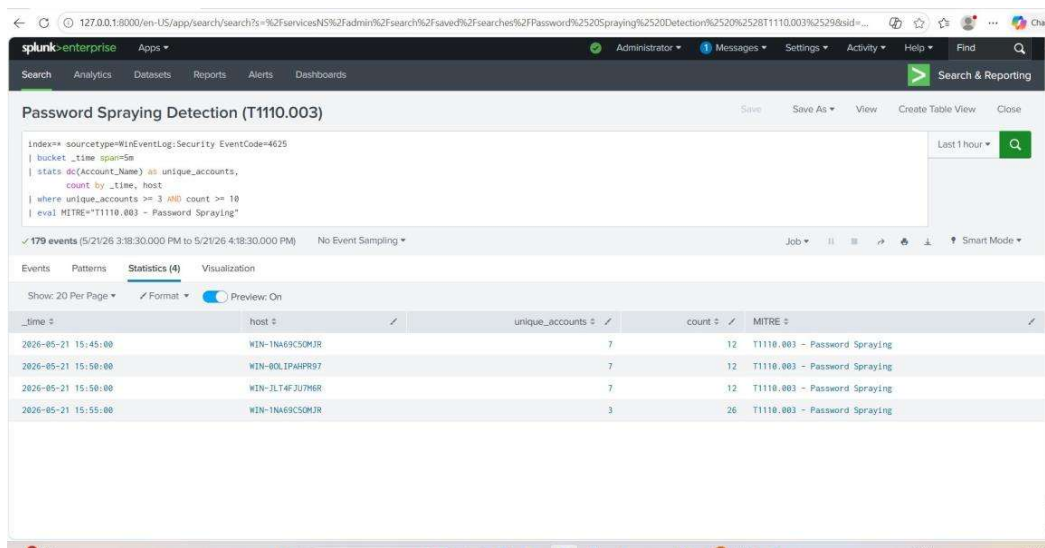


Figure 21: Dashboard — MITRE T1110.003 password spraying detections with severity

7.4 Dashboard — Enhanced Detection with Severity Scoring

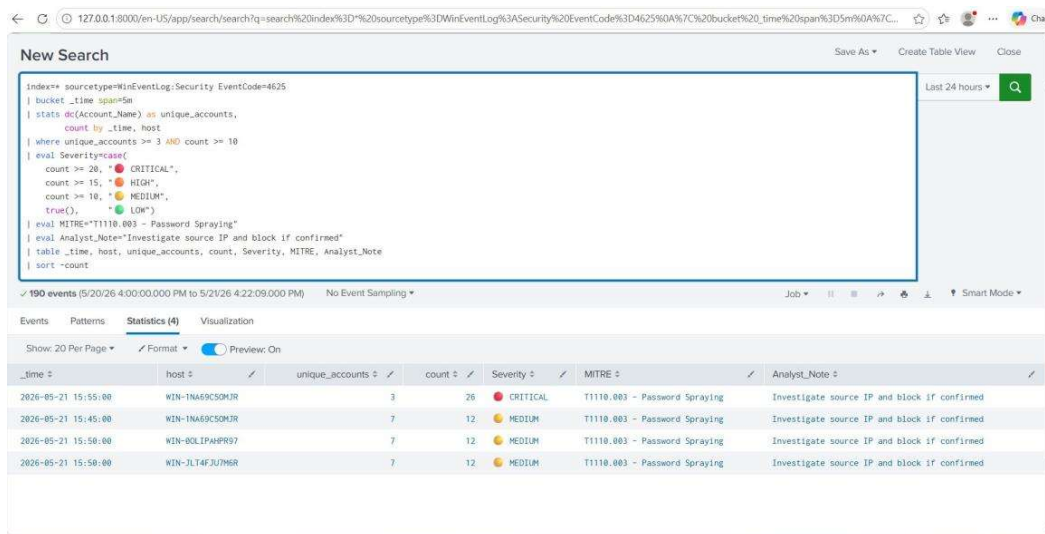


Figure 22: Enhanced T1110.003 query with CRITICAL/MEDIUM severity scoring and analyst notes

8. Key Findings

Finding	Detail	Severity
Password Spraying detected	WIN-00LIPAHPR97: 32 attempts, 8 accounts in 10-min window	CRITICAL
Password Spraying detected	WIN-1NA69C5OMJR: 26 attempts, 3 accounts in 5-min window	CRITICAL
Brute Force confirmed	Administrator account targeted 150 times across all hosts	HIGH
Anonymous logons	167 failed logons from null/anonymous account — network-based attack	HIGH
No compromise confirmed	No EventCode 4624 for targeted accounts post-attack	CLEAR
All forwarders active	24,126 events ingested across 4 hosts and 8 sourcetypes	INFO

9. Challenges & Resolutions

Challenge	Root Cause	Resolution
Forwarder not connecting	Wrong IP 10.148.130.213 in outputs.conf	Updated to correct IP 192.168.1.67
Ping — General failure	VirtualBox set to NAT mode	Changed to Bridged Adapter
Ping — Request timed out	Host firewall blocking ICMP	Added inbound ICMPv4 firewall rule
splunk command not found	Running from C:\Users\Administrator	Navigated to SplunkUniversalForwarder\bin
Login failed on CLI	Used literal text yourpassword	Used actual admin password set at install
Triggered Alerts empty	Severity filter set to High only	Changed filter to All Severities
Total events trellis error	Panel using trellis layout — not PDF-compatible	Disabled trellis in Visualization Format
Event Volume log scale	Chart defaulted to log scale	Changed Y-axis to linear in Format settings

10. Splunk SPL Query Reference

All events by host and sourcetype:

```
index=* | stats count by host, sourcetype
```

Failed login summary (T1110):

```
index=* sourcetype=WinEventLog:Security EventCode=4625  
| stats count by host, Account_Name | sort -count
```

Password spraying detection (T1110.003):

```
index=* sourcetype=WinEventLog:Security EventCode=4625  
| bucket _time span=5m  
| stats dc(Account_Name) as unique_accounts, count by _time, host  
| where unique_accounts >= 3 AND count >= 10
```

Failed logins — 10-minute burst detection:

```
index=* sourcetype=WinEventLog:Security EventCode=4625  
| bucket _time span=10m  
| stats count as Failed_Attempts, dc(Account_Name) as Unique_Accounts by _time, host  
| eval  
Severity=case(Failed_Attempts>=20,"CRITICAL",Failed_Attempts>=10,"HIGH",Failed_Attempts>=5,"MEDIUM",true(),"LOW")  
| eval MITRE=case(Unique_Accounts>=3,"T1110.003 - Password Spraying",true(),"T1110.001 - Password Guessing")
```

Post-attack successful login check:

```
index=* sourcetype=WinEventLog:Security EventCode=4624  
| where Account_Name!="SYSTEM" AND NOT match(Account_Name,"\\\$")  
| table _time, host, Account_Name, Logon_Type | sort -_time
```

Event timeline by host:

```
index=* | timechart span=1h count by host
```


11. Conclusion

This SOC home lab project successfully delivered a complete, end-to-end threat detection and monitoring solution using Splunk Enterprise and the MITRE ATT&CK framework. All primary objectives were achieved:

- 4 hosts connected and forwarding logs to Splunk Enterprise via Universal Forwarders
- 24,126+ events ingested across Windows Event Logs and Performance Monitor sourcetypes
- 3 MITRE ATT&CK-aligned detection alerts created and enabled (T1110, T1110.003, T1110.004)
- Password spraying attacks successfully detected across 3 VMs simultaneously — CRITICAL severity
- 32 failed attempts against 8 unique accounts in a single 10-minute window confirmed detected
- No post-attack credential compromise confirmed through EventCode 4624 analysis
- 8-panel live SOC dashboard built with auto-refresh and email alerting
- Full project documentation produced with screenshots, SPL queries, and MITRE mapping

The skills demonstrated in this project — SIEM configuration, log forwarding, threat detection rule development, attack simulation, incident investigation, and dashboard design — are directly applicable to real-world SOC analyst, blue team, and security engineering roles.

Project Completion Summary

MITRE ATT&CK SOC Home Lab | Rukayat Mopelola Lawal |

Tools: Splunk Enterprise | VirtualBox | Windows Server 2022 | PowerShell

24,126 events | 4 hosts | 3 alerts | 8 dashboard panels | CRITICAL detections confirmed